MILITARY SPECIFICATION

AMPLIFIER, RADIO FREQUENCY
AM-7177A/ARC

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for AM-7177A/ARC Radio Frequency Amplifier hereinafter referred to as the amplifier. The AM-7177A/ARC is a linear amplifier, operating in the 225.0 MHz through 400 MHz band. With an input of nominally 2 watts, the amplifier, with its self-contained automatic level control, provides an output at 30 watt RF level.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 5821

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SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-M-38510	Microcircuits General Specification for
MIL-C-38999	Connector, Electrical, Circular, Miniature, High Density Quick Disconnect, (Bayonet, Threaded, and Breech Coupling) Environmental Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-P-55110	Printed Wiring Boards

STANDARDS

Federal

FED-STD-595	Federal Standard Colors

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Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Electronic Equipment, General Requirements for
MIL-STD-461	Electromagnetic Emission Susceptibiilty Measurements for the Control of Electromagnetic Interference
MIL-STD-462 and Notice 4	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria for Military System Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2084	Maintainability of Avionics and Electronic Systems and Equipment, General Requirements for

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. The issues shall be those in effect on the date of the solicitation.

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DOCUMENTS

Naval Air Systems Command.

AD-1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment
WS-6536	Process Specification Procedures and Requirements for Preparations and Soldering of Electrical Connections

Department of the Navy

NAVMAT P-4855.1	Navy Power Supply Reliability
NAVMAT P-9492	Navy Manufacturing Screening Program

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The radio frequency amplifier covered in this specification shall provide amplification of any frequency in the 225.0 MHz to 400 MHz band from a 2 watt input level to a 30 watt output level. The amplifier shall accommodate either amplitude or frequency modulated input signals and provide comparable output signal modulation. The amplifier shall be broadband, requiring no tuning time for change from one input frequency to any other frequency. The amplifier along with a receiver-transmitter (RT), and filter make up a system (EMC Filter/HPA System).

3.1.1 Module complement. The amplifier shall consist of the following major shop replaceable subassemblies (SRAs):

a. A1 RF Amplifier

b. A2 Power Supply/ALC

Each of these major modules may include subassemblies for providing manufacturing and maintenance benefits; however they are not required to be directly plug-in or replaceable without final adjustment after installation. Final adjustment of the automatic level control is also permissible after installation of the A1 and A2 modules into the amplifier.

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3.2 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Parts and materials. In the selection of parts and materials, the prime considerations are fulfillment of performance requirements, logistics supportability and reliability assurance in a cost effective design. The following shall govern in parts and material selection.

a. Subassemblies shall be designed and fabricated to be repairable. Maximum economic standardization of parts and materials shall be exercised. Parts selection shall be made in the following order of precedence:

(1) Military Standard parts shall be selected in accordance with MIL-STD-454, Requirement 22.

(2) Nonstandard parts previously qualified for other programs.

(3) Nonstandard parts.

b. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similiar military standard or standard nonestablished reliability part. Integrated circuits and hybrids shall use MIL-STD-883, Method 5004, Class B, as a basis for tailored screens.

3.3.1 Component rescreening. The contractor shall conduct 100 percent incoming inspection of all active components used in production equipment and spares. Functional and parametric checks of active components will be performed at the components rated temperature extremes and at ambient. Components which are not compliant with their specification shall not be used in the ARC-182.

3.3.2 Parts derating and application. All parts used shall be applied well within their ratings. The derating shall encompass the appropriate and meaningful application conditions such as voltage, current, power, temperature, mechanical, and duty cycle. Electronic and electromechanical parts shall conform to Table I electronic parts derating for worst case electrical and environmental stress unless formal written approval is received from the procuring activity prior to incorporation into the design. Part level stress analysis shall be used to verify that all parameter stresses are within the derated values at worst case circuit and environmental conditions. The following limitations on parts usage shall apply:

a. Prohibited application

(1) Non-hermetically sealed semiconductors except power transistors
(2) Photo couplers (photo transistors)
(3) Hot carrier (Schottky) power rectifiers
(4) Microcircuit sockets
(5) Non-hermetically sealed wet tantalum capacitors

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TABLE I. Electronic parts derating.

Part Type	Parameter	Maximum Allowable Application Stress
Capacitor	Applied Voltage (Ripple Voltage for Tantalum)	50% of rating 1/
Resistors	Power Voltage	50% of rating 70% of rating
Diodes	Power Current Voltage Junction Temperature	50% of rating 75% of rating 60% of rating 2/
Transistors	Power Voltage Junction Temperature	55% of rating 60% of rating 2/
Integrated Circuits		
Digital	Fan-Out, Fan-In Junction Temperature	80% of rating 2/
Linear	Power Junction Temperature	75% of rating 2/
Transformers, Inductive Devices	Parts shall be used at a winding temperature of at least 30 degrees below rated temperature. In addition, the maximum voltage between windings or between a winding and a shield shall be not greater than 60% of rated voltage for the steady state conditions and 90% of rated voltage with transients. General purpose inductors, audio and power transformers shall be applied at no more than 70% of rated current.	

1/ For solid tantalum and all electrolytic capacitors, the maximum applied voltage shall be not greater than 70 percent of the vendor rated value. "Applied voltage" is the sum of the applied peak ripple voltage and the applied DC voltage. High Q ceramic capacitors, the applied voltage shall be not greater than 80 percent of the vendor rated value.

2/ Semiconductor devices shall be mounted and operated in such manner to insure worst-case junction temperature does not exceed 110 degrees C at +71 degrees C ambient temperature. Any deviation requires procuring activity approval. Power semiconductor devices, on a very selective basis, may be mounted and/or operated with worst case junction temperatures not to exceed 150 degrees C at +71 degrees C ambient temperatures.

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3.3.3 Parts control program. The contractor shall establish and maintain a parts control program in accordance with the requirements of MIL-STD-965, Procedure I and as specified herein. The program shall control part standardization and the criteria for the selection, application, and testing of parts to achieve the reliability required. As a minimum, a parts program equivalent to that specified herein shall be required of subcontractors and suppliers.

3.3.4 Non-standard parts approval. The contractor shall obtain procuring activity approval of all non-standard parts used in the equipment. Standard parts are defined in MIL-E-5400. In addition to the MIL-E-5400 requirements for standard parts, microelectronic parts are standard only if:

a. The device is listed in MIL-STD-1562, and

b. Sources (at least two) are listed on the Qualified Parts List (QPL), and

c. The part is procured from one of the qualified sources, and

d. Quality level "B" (or better) of MIL-M-38510 is required.

The approval procedure shall be in accordance with MIL-E-5400 and MIL-STD-965, Procedure I. In addition to the requirements of MIL-E-5400, approval of non-standard parts will be based on:

a. Suitability for the application

b. Conformance to this specification and to MIL-E-5400

c. Interchangeability

d. Screening (burn-in) method

3.3.5 Samples of non-standard parts. Samples of non-standard parts are not required unless specified in the contract. Where non-standard parts samples are specified in the contract, the sample shall be in accordance with MIL-E-5400 (see 6.2.1).

3.3.6 Cabling and connections.

3.3.6.1 Cables and connectors. The amplifier shall provide for the use of cables and connectors in accordance with MIL-E-5400. The design of the connectors shall be such that improper assembly, mating or installation is impossible.

3.3.6.2 Interconnection cabling. The amplifier shall be capable of required operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provide the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the amplifier.

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3.3.7 Modules. The electronic portions of the amplifier shall be functionally modularized in accordance with MIL-STD-2084. Conformal coatings, encapsulants, embedments or potting material used with repairable modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.3.8 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of the amplifier in accordance with MIL-E-5400. The amplifier shall be constructed to preclude the requirement for any adjustment when replacing an item by another item of the same type designation.

3.4 Design and construction. The amplifier shall conform with all the applicable requirements of MIL-E-5400 for design and construction except where otherwise specified herein. In any case, it is a condition of final acceptance that the amplifier shall meet all the performance requirements of this specification.

3.4.1 Reliability. Unless otherwise specified in the contract or purchase order, the contractor shall establish a reliability program conforming to MIL-STD-785 (see 6.2.1). Task 104 of MIL-STD-785 shall be implemented to establish and maintain a closed-loop failure reporting system. This system will identify failure trends and patterns at all levels of assembly prior to acceptance of the hardware and will also identify failure trends related to field failures throughout the contract period. The contractor shall develop and implement corrective action for significant failure trends. The reliability program shall include provisions for assuring the reliability of the amplifier or any portion of the equipment when supplied for use as spare or repair part.

3.4.1.1 Operational stability. The amplifier shall operate with specified performance for at least the specified mean (operating) time between failures without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The amplifier shall have a total operating life of 20,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.1.3 Reliability in MTBF (mean time between failure). The amplifier, including, any built-in-test (BIT) provisions, shall have a specified mean (operating) time between failures, upper test MTBF (θ_0) of 2000 hours (as specified in MIL-STD-781), and a lower test MTBF (θ_1) of 1000 hours when tested and accepted as specified under the requirements of 4.4.5 and 4.5.4.

3.4.2 Maintainability. Maintainability (built-in test features, construction and packaging, provision for test points and other maintainability parameters) shall be as specified herein with MIL-STD-2084 being used as a guide. The fault detection percentage requirement of MIL-STD-2084 shall be 75 percent minimum of the critical path circuitry. The amplifier shall be considered a WRA; all packages within the amplifier, including chassis and wiring, shall be considered SRA's.

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3.4.2.1 Compatibility with ATE. The amplifier shall be compatible with Automatic Test Equipment (ATE) for fault verification.

3.4.2.2 Maintainability design. All electrical or electronic circuits and parts shall be packaged on replaceable and repairable, plug-in modules. Circuit breakers, indicators, and other items which may logically be bolted and soldered, or otherwise more rigidly electrically and mechanically fastened to the amplifier housing shall not be considered modules. SRAs (Shop Replacement Assemblies) shall be QRAs (Quick Replaceable Assemblies) as specified in MIL-STD-2076. SRA arrangement shall be such that access to any SRA does not require the removal of adjacent SRAs or parts other than access panels. The use of a special tool or tools shall not be required for SRA removal. Fuses may be used in lieu of circuit breakers.

3.4.2.2.1 Keying. Connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the amplifier.

3.4.2.2.2 Access panels. Access panels shall be retained by fasteners suitable to the procuring agency.

3.4.2.2.3 Service access. Fasteners required for removal of the WRA shall be held by retaining devices suitable to the procuring agency.

3.4.2.2.4 Adjustments. No adjustment or alignment shall be required at the organizational level. Adjustment or alignment if required at a higher maintenance level (intermediate or depot), shall not require the removal of the part to be adjusted or aligned, or of the hardware element to which it is attached.

3.4.2.2.5 Test points. All test points shall be as specified herein to accommodate approved test equipment. MIL-STD-2084 takes precedence over MIL-STD-415 requirements. The sub-SRA test points and related connector requirements of MIL-STD-2084 are waived where prohibited by space or circuit performance considerations. The sub-SRA test points and related connector requirements philosophy shall be reflected in appropriate maintenance data. MIL-STD-2084 and MIL-STD-415 shall be used as a guide.

3.4.2.2.5.1 Module test points. Test points shall become accessible with minimum removal of protective covers or enclosures.

3.4.2.2.5.2 Test points utilized at intermediate or depot level. Test points shall be provided for verification of faulty equipments at the intermediate level and repair of equipments and modules at the depot level.

3.4.2.2.5.2.1 Power supply test points. Accessible, individual test points shall be provided on the amplifier to measure the power supply voltages. The power supply test points and related connector requirements philosophy shall be reflected in appropriate maintenance data.

3.4.2.2.6 Organizational level maintainability. The amplifier shall provide the operator or organizational maintenance technician, while airborne or on the ground with performance/readiness test and fault (GO/NO-GO) indication.

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3.4.2.2.6.1 Performance/readiness test requirements. BIT provisions shall be incorporated in the amplifier to provide a GO/NO-GO indication of amplifier readiness when used in conjunction with operational tests. These provisions may be in conjunction with, but not necessarily as part of, the fault isolation features. The operation of the performance/readiness test provision shall be both a manually energized or continuous test mode of operation. Normal equipment performance may be interrupted during the manually initiated BIT sequence and, if necessary, switches may be used in series paths for purposes of introducing BIT stimuli or measuring equipment performance. Self-test of the BIT is not required.

3.4.2.2.6.2 Fault isolation. In conjunction with, but not necessarily as part of, the performance/readiness test features, there shall be features incorporated which provide fault isolation to the WRA.

3.4.2.2.6.3 MTTR (mean time to repair). For organizational maintenance, the MTTR shall be not greater than 15 minutes to remove and replace and verify operation of the replaced equipment. The amplifier design shall be such that no ancillary test equipment is required for organizational level maintenance.

3.4.2.2.7 Intermediate level maintainability. The time to verify performance at the intermediate level maintenance shall be not greater than 20 minutes.

3.4.3 Transportability. The amplifier shall be transportable by air, rail, truck and ship when packaged as specified in Section 5.

3.4.4 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests, except that required field tests may be conducted under outside ambient conditions:

a.	Temperature	Room ambient
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power voltage	28 $\pm$ 0.5 volt dc

3.4.4.1 Service conditions. The amplifier shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 1 equipment.

3.4.4.1.1 Sinusoidal vibration. The amplifier shall operate as required when subjected to a sinusoidal vibration test in accordance with MIL-T-5422 when operated in a system configuration on the MT-6330/ARC Mount. The amplifier shall be tested to Curve IIA of MIL-E-5400 with mount isolators removed (hard mounted to the vibration table) and to Curve IVA of MIL-E-5400 with mount vibration isolators in place.

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3.4.4.1.2 Random vibration. The amplifier shall operate as required when subjected to the random vibration requirements specified in 4.4.4 and mounted on the MT-6330/ARC Mount.

3.4.4.1.3 Thermal conditions. The amplifier shall operate as required under the thermal conditions specified by MIL-E-5400, Class 1, when installed in the MT-6330/ARC with blower operational or with equivalent forced air cooling.

3.4.4.1.4 Temperature shock. The amplifier shall meet the temperature shock requirements of MIL-T-5422.

3.4.4.2 Warm-up time. The time required for the amplifier to warm up prior to operation shall be kept to a minimum and shall be not greater than 10 seconds at standard conditions. Under service conditions, the warm-up time shall be not greater than 20 seconds.

3.4.4.3 Electromagnetic interference (EMI). The amplifier shall comply with the requirements of MIL-STD-461 with for Class A1b equipment except as modified herein. These requirements shall be met in all operating modes and at all operating frequencies of the amplifier. MIL-STD-461, Part 1 shall be modified by Tables II, III and the following additions and deletions:

a. Additions

(1) NAVAIR AD-1115

b. Deletions

(1) AFSC DH 2-5
(2) AFSC DH 2-7
(3) AMC Pamphlet 706,235
(4) NAVELEX 01001,106

3.4.4.3.1 Conducted and radiated emissions limits. The conducted and radiated emissions of the amplifier shall comply with the limits of the following requirements of MIL-STD-461, Part 2, except the RE02 transmit frequency may be up to 35 dB above the limit, and harmonics may be up to 15 dB above the limit:

a. CE01

b. CE03

c. CE07

d. RE01

e. RE02

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TABLE II. RS03 signals

Frequency (MHz)	Modulation				Power Density (mW/cm^2) Peak 5/	Avg	v/m (ref) 6/
0.014-8507/	AM 90% 1 kHz Sine Wave				-	.106	20
215-225	PAM1/	20 usec2/	PW3/	200 pps4/	.317	-	35
215-225	PAM	200 usec	PW	200 pps	.317	-	35
400-450	PAM	15 usec	PW	300 pps	.106	-	20
850-940	PAM	10 usec	PW	1000 pps	.106	-	20
850-940	PAM	125 usec	PW	300 pps	.106	-	20
1215-1365	PAM	10 usec	PW	3000 pps	.106	-	20
1365-2900	PAM	1 usec	PW	1000 pps	.106	-	20
2900-3100	PAM	3 usec	PW	1000 pps	1.6	-	78
2900-3100	PAM	70 usec	PW	200 pps	1.6	-	78
3100-9000	PAM	1 usec	PW	1000 pps	.106	-	20
9000-14000	PAM	0.2 usec	PW	1000 pps	.106	-	20
14000-18000	PAM	0.2 usec	PW	50000 pps	.106	-	20

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and/or field strength shown.

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TABLE III. High level RS03 signals

Frequency (MHz)	Modulation				Power Density (mw/cm^2) 5/	v/m (ref) 6/	Scan Rate (sec) 7/	Dwell Time (msec) 7/
220 8/	PAM1/	10 usec2/	PW3/	200 pps4/	10.6	200	4	500
220	PAM	200 usec	PW	200 pps	10.6	200	4	500
425	PAM	15 usec	PW	300 pps	10.6	200	10	200
900	PAM	2 usec	PW	1000 pps	10.6	200	5	50
900	PAM	125 usec	PW	300 pps	10.6	200	5	50
1250	PAM	10 usec	PW	3000 pps	10.6	200	5	50
1350	PAM	5 usec	PW	3000 pps	10.6	200	5	50
3000	PAM	3 usec	PW	1000 pps	10.6	200	5	50
3000	PAM	70 usec	PW	200 pps	10.6	200	5	50
3600	PAM	1 usec	PW	1000 pps	10.6	200	5	50
5600	PAM	1 usec	PW	600 pps	10.6	200	5	50
9100	PAM	0.5 usec	PW	1000 pps	10.6	200	5	50
9375	PAM	0.2 usec	PW	1000 pps	10.6	200	5	50
13900	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
15500	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
33200	PAM	0.2 usec	PW	2000 pps	10.6	200	0	CONT

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is defined as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Scan and Dwell refer to simulated antenna rotation. The signal shall be disabled except for the specified dwell. The signal shall be periodically enabled for the specified dwell time with the period equal to the specified scan time.
8/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations shown.

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3.4.4.3.2 Conducted and radiated susceptibility limits. The amplifier shall comply with conducted and radiated susceptibility limits of the following requirements of MIL-STD-461:

a. CS01

b. CS02

c. CS06

d. RS01

e. RS02

f. RS03

The amplifier shall exhibit no susceptibility as specified in 3.3.4.3.3 when subjected to an RS03 test to the limits of Table II. In addition, the amplifier shall exhibit no susceptibility in accordance with the criteria of 3.4.4.3.4b when subjected to signals at the levels of Table III.

3.4.4.3.3 Susceptibility. The following criteria shall be applied to determine whether the amplifier is susceptible:

a. Any failure of the amplifier to comply with the requirements of 3.6.1, 3.6.3, 3.6.6, and 3.6.7 shall be considered an indication of susceptibility.

b. Any repeatable non-commanded change in the modes of operation of the amplifier shall be considered an indication of susceptibility. This shall apply whether the failure is transient (momentary) or non-transient (permanent) in nature. In addition, the inability to change the mode of operation while exposed to the susceptibility signal shall be considered an indication of susceptibility.

3.4.4.3.4 Ground plane interference. All amplifier interface circuits shall be capable of specified performance when subjected to the following aircraft chassis noise between the WRA's.

a. Three volts rms from 320 Hz to 500 Hz. (Not to be greater than 150ma rms applied current.)

b. One volt rms from 500 Hz to 20 MHz. (Not greater than 150ma rms applied current for frequencies between 500 Hz and 50 kHz. Not to be greater than 1 watt (CW) from a 50-ohm source for frequencies above 50 kHz.)

c. ± eight volt pulses, 100 usec wide at 100 pps. (Not to be greater than 15a peak current spikes.)

In addition to the susceptibility specified in 3.4.4.3.3, the radio shall maintain intelligible communications during a "talk thru" test with another radio set using a low level signal (not to exceed 20 microvolts input to the receiver). The test will be performed with a test set up employing a RT-1250A/ARC, RT-1324A/ARC, or equivalent receiver-transmitter.

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3.4.4.3.5 Conducted susceptibility test, interconnecting and signal leads. The amplifier shall exhibit no susceptibility, as specified in 3.4.4.3.3 when subjected to 100 mv rms 2 to 30 MHz coupled into its interconnecting and signal leads.

3.4.4.3.6 Grounding requirements. The grounding system between WRA's of the equipment shall be compatible and consistent with the aircraft grounding scheme specified in 3.4.4.3.6.1 through 3.4.4.3.6.3. All WRA's shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, provide effective shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.4.4.3.6.1 Chassis grounds. A ground wire of minimum length connected internally to the WRA chassis shall be provided at a pin on each primary power connector. No circuit shall be allowed to utilize this wire as its primary return.

3.4.4.3.6.2 Shielding grounding. A separate connector pin shall be provided for each aircraft wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the equipment chassis inside the WRA adjacent to the connector mounting and by the shortest means practical.

3.4.4.3.6.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, and bushings shall be grounded to the chassis.

3.4.4.3.8 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

3.4.5 Dimensions. The dimensions of the amplifier shall be not greater than the following (see Figure 1):

a. Height - 5.6 inches

b. Width - 4.6 inches

c. Length - 5.0 inches

3.4.6 Weight. The weight of the amplifier shall be not greater than 4.2 pounds.

3.4.7 Color. The exterior color of the amplifier shall be lusterless black. Painted surfaces shall be in accordance with FED-STD-595, Color No. 37038 and anodized surfaces shall meet the requirements of MIL-A-8625 Type II, Class 2, Color Black.

3.4.8 Nameplates and identification marking. Nameplate approval and identification marking shall be in accordance with MIL-N-18307.

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3.4.8.1 Decals, nameplates, and marking locations. All decals, nameplates, and markings shall be affixed to the amplifier as shown on Figure 1.

3.4.8.2 Serial numbers. Serial numbers will be assigned by the procuring activity for the amplifiers under contract or purchase order. The contractor shall serialize sequentially all amplifiers which are delivered under contract.

3.4.9 Input electrical power.

3.4.9.1 Operating power. The amplifier shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The voltage supplied shall be 28 volts dc. The power required shall not be greater than 280 watts.

3.4.9.2 Degraded performance. Degraded performance will be permitted for voltage transients with durations of 75 microseconds to 0.5 second during normal electrical system operation. Operation shall return to normal within five seconds with no resulting damage to the amplifier.

3.4.9.3 Emergency voltage. During operation in the emergency steady state electric power condition as specified in MIL-STD-704, the amplifier shall remain safe. The emergency power source will impose an interval of zero voltage at time of loss of primary power for a maximum period of 30 seconds. The amplifier is not required to perform during this period of power loss but shall return to performance as specified above after return to emergency power conditions in accordance with MIL-STD-704 and following the first new instruction from the control system. RF power output at emergency limit voltage shall be six watts minimum.

3.4.9.4 Undervoltage protection. The amplifier shall not be damaged by voltages below the minimums specified by MIL-STD-704, and shall be capable of resuming normal operation when the voltage returns within normal limits.

3.4.9.5 Reverse polarity protection. The amplifier shall be protected such that it will not be damaged by application of reverse polarity voltage.

3.4.10 Manufacturing requirements. The amplifier shall be manufactured in accordance with the manufacturing processes, workmanship, and quality assurance provisions of MIL-STD-454, MIL-STD-275, MIL-P-55110, WS-6536, NAVMAT P-4855.1, and NAVMAT P-9492.

3.5 Performance. Unless otherwise specified herein, values set forth to establish specified performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein.

3.5.1 Operation. The amplifier shall provide the functions and capabilities specified herein for operation in military aircraft and helicopters. Additionally the amplifier shall be compatible with associated equipment listed in 6.4 to provide operations specified in 3.6.

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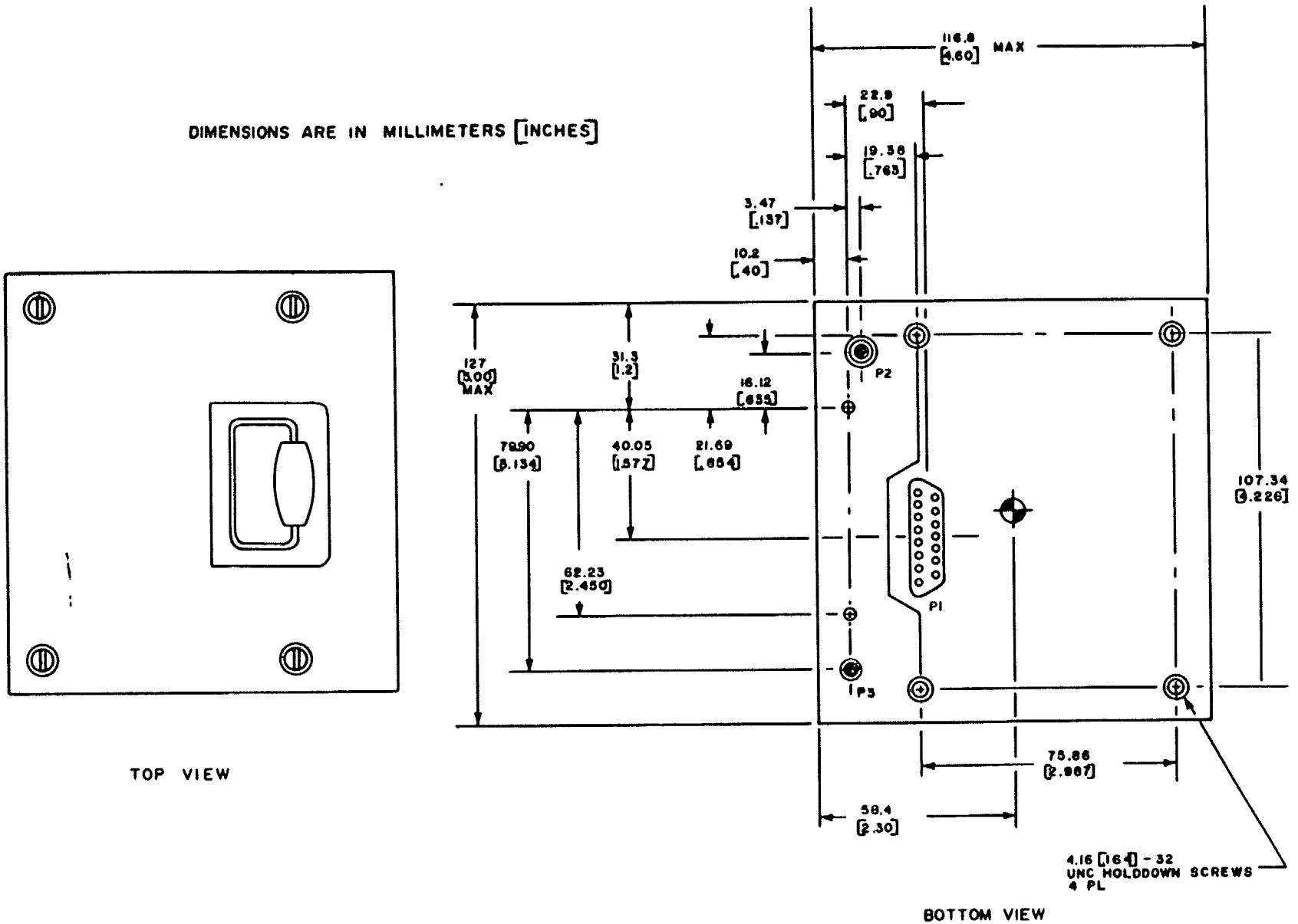


FIGURE 1. Outline dimensions for Am-7177A/ARC Radio Frequency Amplifier

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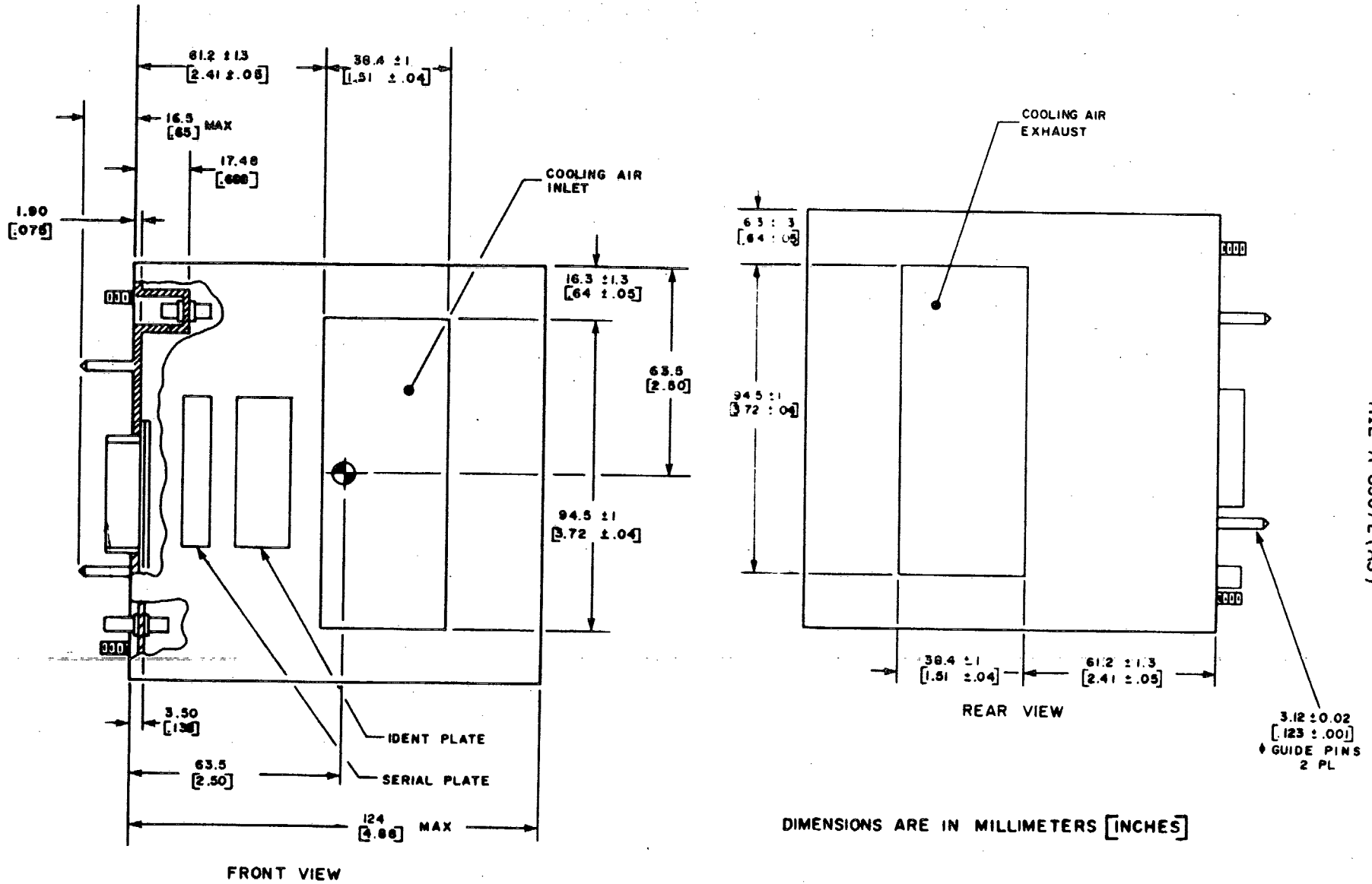


FIGURE 1. Outline dimensions for Am-7177A/ARC Radio Frequency Amplifier - continued

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3.6 Detail requirements. The amplifier shall provide an independent function, however, it is normally attached to the MT-6330/ARC Mounting and will be used with the RT-1250A/ARC receiver-transmitter and UHF Bandpass Filter, F-1556/ARC.

3.6.1 Function. With an input of nominally two watts, the amplifier with its self contained automatic level control (ALC) shall provide an output at the 30 watt RF level.

3.6.1.1 Physical configuration. The amplifier configuration shall conform to Figure 1. Dimension shall be as specified in 3.4.5. The amplifier in addition to electronics function shall have the following configuration provisions:

a. Hold down screws for securing to a chassis.

b. Power and control wiring connector and input and coaxial connectors.

c. Air flow provisions for directing cooling air to heat dissipating components.

3.6.2 Controls. The amplifier shall contain control circuitry for providing specified output signal level when minimum specified input signal level is present. This circuitry shall also provide protective power turn down control when operative temperature becomes excessive, and provide protection from external adverse mismatch of the RF output impedance loading. The amplifier shall properly respond to transmitter keying, and transmit blanking control signals.

3.6.3 Operational modes. The operational modes shall be standby and transmit. The amplifier shall be controlled by the amplifier excitation input. The transmit blanking shall inhibit the amplifier.

3.6.4 Duty cycle. With cooling air as provided by the MT-6330/ARC or equal, the amplifier shall operate as required over the MIL-E-5400, Class 1, temperature altitude environmental range with a duty cycle one minute transmit and five minutes standby continuous operation basis.

3.6.5 Electrical connections. Connections to external circuits shall be as specified in Table IV:

TABLE IV. Electrical connections

Reference Designation	Receptacle Type	Function
P1	M24308/3-2	Power/Control
P2	PN 357-7207-290	RF Input
P3	PN 357-7207-290	RF Output

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3.6.5.1 Power/function control wires. The amplifier shall interface with the MT-6330/ARC Mount P1 connector pin designations and functions as specified in Table V.

3.6.5.2 RF input. The RF input circuit shall have a nominal 50 ohm impedance and shall be compatible with input power level of 1.5 watts to 3.0 watts. The RF input circuit shall have a voltage standing wave ratio (VSWR) of 1.5:1 or lower.

3.6.5.3 RF output. The RF output circuit shall be compatible with 50 ohm external load impedance. Performance versus loading shall be as specified in 3.6.6.2 for low and high VSWR conditions.

3.6.6 RF output characteristics. The RF output characteristics versus input parameters, thermal conditions, output loading and primary power shall be as specified in 3.6.6.1 through 3.6.6.7.

3.6.6.1 RF power output - normal conditions. With an input RF power of 1.5 to 3.0 watts, the output power shall be not less than 27.5 watts and not greater than 37.5 watts.

3.6.6.2 RF power output with high VSWR. The output power of the amplifier shall automatically reduce as the output voltage standing wave ratio (VSWR) increases above 2.5:1. However, at infinite VSWR the output shall be not less than 6 watts.

3.6.6.3 Thermal protection characteristics. The amplifier shall contain self protection circuitry to prevent damage from overheating. The system shall gradually turn the power output down, but retaining an output of at least six watts, to prevent the heat sink temperature from rising above 100°C.

3.6.6.4 RF output with primary power variation. The output power level stated in 3.6.6.1 shall be required with an input voltage of 26 volts dc or higher. For voltages from 26 volts dc to 18 volts dc, the output may be gradually reduced to six watts minimum with 18 volts dc input.

3.6.6.5 Modulation characteristics. The amplifier shall provide nominal linear amplification of the incoming signal, thus having a minimal effect on modulation characteristics. The modulation distortion shall not be increased more than ten percent from that of the incoming signal. The output average modulation shall be within ten percent of the incoming signal modulation level for 1 kHz to 10 kHz modulation.

3.6.6.6 Harmonic emission, spurious frequencies. The harmonic emission of the amplifier shall be not greater than a level of 60 dB below the carrier level. The spurious frequency levels of the output signal shall be not greater than 10dB above the incoming signal relative to the respective carrier levels. Spurious frequencies initiated within the amplifier shall be not less than 80 dB below the carrier level.

3.6.6.7 RF amplifier noise level (noise floor). With an input signal having less than -150 dBm/Hz broadband noise content, the amplifier output noise floor at ± 5 percent from the carrier shall be -140 dBm/Hz or lower.

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TABLE V. Amplifier power/control connector P1 interface characteristics

Pin No.	I/O	Function	Characteristics
1	I	+28 volts No. 1	Primary Power in accordance with MIL-STD-704, 4 amperes maximum
2	I	+28 volts No. 1	Primary Power in accordance with MIL-STD-704, 4 amperes maximum
3	-	Ground, Power Return	Internally connected to pins 10 and 11
4	-	Ground, Power Return	Internally connected to pins 10 and 11
5	-	Spare	
6	I	Amplifier Key	0 ± 0.5V dc causes ALC precharge circuit disable
7	0	Amplifier Fault	0 to 1.5V = No fault 3.5 to 5.0V or open = fault
8	I	XMIT BLANKING	0 ± 0.5V dc causes ALC precharge circuit enable
9	I	+28 volts No. 2	Primary Power in accordance with MIL-STD-704, 2 amperes maximum
10	-	Ground, Power Return	Internally connected to pins 3 and 4
11	-	Ground, Power Return	Internally connected to pins 3 and 4
12	-	Digital/Signal Ground	Signal ground reference
13		Antenna Fault	0 to 1.5V is no fault, open or 3.5 to +5 volts is fault designation 100 MA capability
14	-	Spare	
15	0	Sidetone	0.56 ± 0.15 VRMS out for 30 watt RF carrier modulated 30% at 1000 Hz

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3.6.7 Amplifier turn-on and turn-off time. The amplifier shall provide an output not less than 50 percent of the stabilized level within 50 milliseconds after the transmit key signal and input driving signal are present. Turn off shall occur within one millisecond after the transmitter key signal and input driving power are removed. The amplifier turn-on and turn-off time shall be compatible with the HAVE QUICK system operation.

3.6.7.1 Transmit blanking. Transmitter blanking shall inhibit transmission.

3.6.8 Fault indication-amplifier. An amplifier fault indication output shall be provided for low RF power, low DC input voltage, and high heatsink temperature. Fault threshold levels shall be:

a.	RF power output	9 3 watts
b.	DC primary power input	18 ± 2 Vdc
c.	Heatsink temperature	99 ± 10°C

3.6.8.1 Fault indication-antenna. The amplifier shall monitor the ratio of forward and reflected RF power and indicate an antenna fault if the resulting VSWR is greater than 5:1.

3.6.9 AM sidetone characteristics. The amplifier shall generate an AM sidetone output by sampling the RF carrier in the UHF band. The output level shall be 0.56 VRMS ± 0.15 VRMS for a 30 watt carrier modulated 30 percent at 1000 Hz.

3.7 Safety. The amplifier safety requirements shall be in accordance with MIL-E-5400 except resistance from ground pin to chassis shall be 20 milliohms maximum.

3.7.1 Human performance/human engineering. MIL-STD-1472 and MIL-H-46855 shall be used as a design guide for the amplifier and all component parts therein.

3.8 Bonding. Bonding shall be in accordance with MIL-B-5087.

3.9 Workmanship. Workmanship for the amplifier shall conform with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Standard conditions. Unless otherwise specified herein, all inspections shall be performed with conditions specified as follows:

a.	Temperature	Room ambient (25 $\pm$ 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	28 $\pm$ 0.5 volt dc

4.4 First article inspection. First article inspection shall be conducted by the contractor on amplifiers representative of the production amplifiers to be supplied under the contract. First article tests shall be accomplished under the authorized test procedures (see 4.6). No first article inspections shall be conducted prior to acceptance of the first article test procedure by the procuring activity. First article inspection shall consist of all the tests described in 4.4.1.

4.4.1 Scope of first article tests. First article tests shall include all tests in the approved test procedures to determine that the amplifier meets all the requirements of this specification, other applicable specifications and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 including temperature shock, a random vibration test as specified in 4.4.4, and an electromagnetic compatibility test in accordance with MIL-STD-461 and MIL-STD-462, as modified in 4.4.3. The first article test may also include a reliability qualification test.

4.4.2 First article approval. Approval of the first article sample will be by the procuring activity upon completion of all tests.

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4.4.3 Electromagnetic interference tests. Compliance with the requirements of 3.4.4.3 and subparagraphs shall be demonstrated by testing in the system configuration. These tests shall be performed in accordance with MIL-STD-462, with Notice 2 as modified herein.

4.4.3.1 EMI test plan. The contractor shall perform EMI tests in accordance with an EMI Test Plan. The Test Plan shall describe all EMI tests required to demonstrate compliance with the requirements of 3.4.4.3, and shall address all anomalies, inconsistencies, and omissions (if any). The Test Plan shall detail a combination of operating modes and frequencies to be tested. The combination shall represent all operating modes of the amplifier and at least three frequencies as specified by 4.4.3.2. The Test Plan shall be acceptable to the procuring activity prior to the commencement of EMI testing.

4.4.3.2 Ground plane interference. The contractor shall prepare a procedure to demonstrate that the amplifier complies with the requirements of 3.4.4.3.4. The procedure shall be included in the EMI Test Plan.

4.4.3.3 Conducted susceptibility test, interconnecting and signal leads. In addition to the requirements of MIL-STD-461 the following RF susceptibility test shall be conducted. No malfunction, undesirable response, or change in indication beyond the tolerance given in the equipment specification or approval test plan shall be produced in any equipment when the RF signal levels listed in Table VI are coupled into the interconnecting wiring.

TABLE VI. Susceptibility test signals

Frequency	Required Signal Levels In One Turn Loop
2 to 30 MHz	100 MV RMS

The RF signal shall be modulated 80 percent with 400 Hz and with that frequency and the type of modulation to which the test sample is most susceptible. This requirement applies to all interconnecting wires or cable between component parts of subsystems, and to wires and cables connected to other equipment. The RF signal shall be measured in the one-turn loop with an RF voltmeter as shown on Figure 2. The signal generator shall have an output impedance of 50 ohms or less and a maximum open-circuit voltage of not less than 3 volts rms. The test procedure shall be as follows:

a. With the current probe clamped around the wire bundles under test, set the signal generator for maximum output and slowly scan from 2 MHz to 30 MHz.

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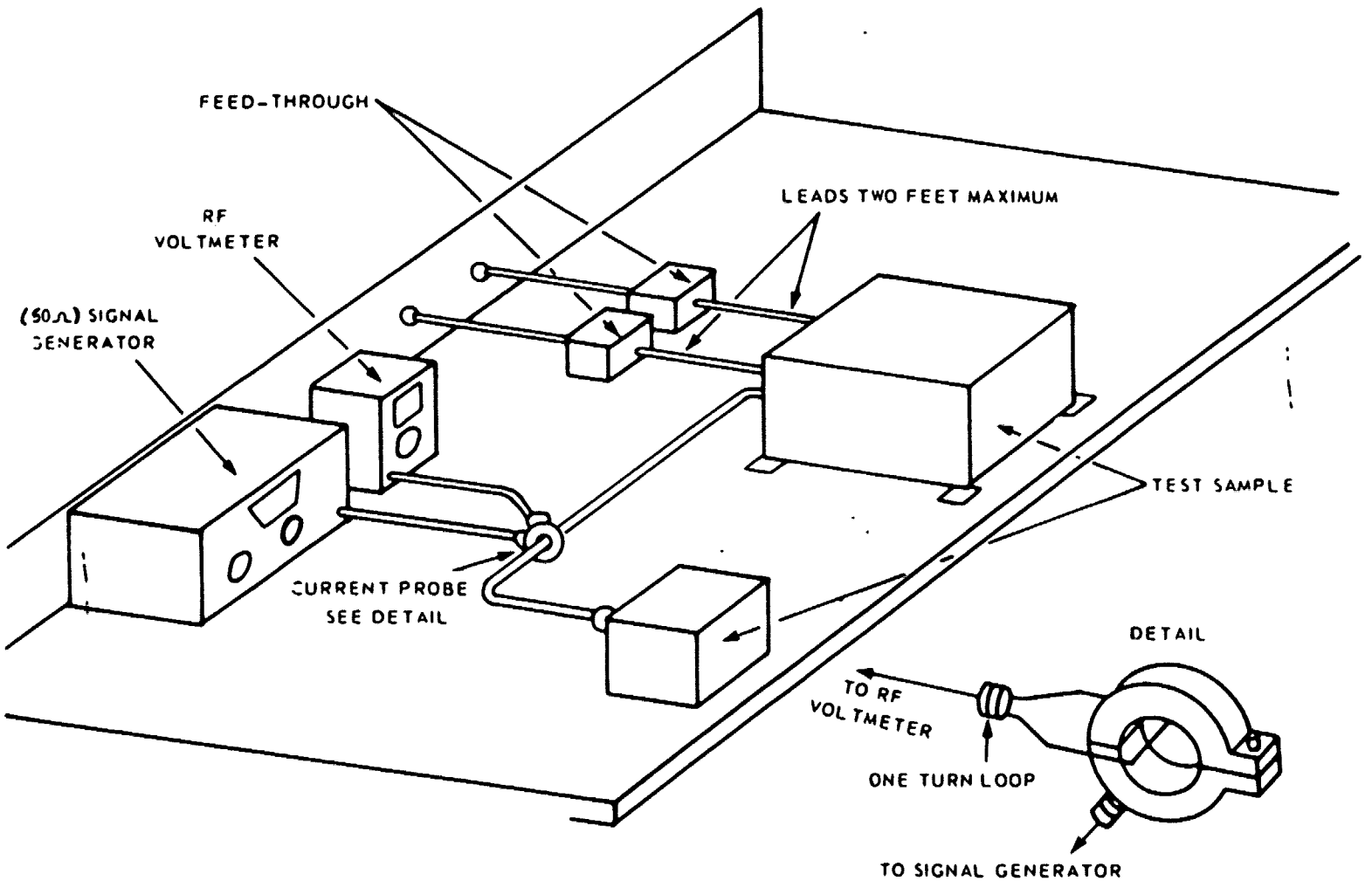


FIGURE 2. Conducted susceptibility test set up

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b. At those frequencies where a response is noted, move the current probe along the wire for maximum indication. Reduce the signal generator output and determine the threshold of susceptibility of the wire bundle. When a response is found on an interconnecting wire bundle, probe each wire in the bundle separately to determine the susceptible wire(s).

c. Record and include in the test report the threshold of susceptibility data (frequency, amplitude, and description of responses) obtained in (b) above. Also include in the test report the corrective action required for each wire that does not meet the required level.

Wire bundles may be tested in sections if the current probe will not fit around the complete wire bundles. Chassis or circuit grounds not normally routed with the interconnecting wiring shall be measured individually and shall not be included in the wire bundle under test. Power ground return leads shall not be placed in the probe.

4.4.3.4 CE03 test procedure. The CE03 requirement of 3.4.4.3.1 shall be demonstrated using MIL-STD-462, Notice 4, Test Method CE03 for all power leads and MIL-STD-462, Notice 4, Test Method CE04 for all control and signal leads tested.

4.4.3.5 CE07 test procedure. The contractor shall prepare a test to demonstrate compliance with the CE07 requirement of 3.4.4.3.1. This test shall be included as a part of the EMI Test Plan.

4.4.4 Random vibration test. A random vibration test shall be conducted as specified in 4.4.4.1 through 4.4.4.4.

4.4.4.1 Test item operation. The amplifier shall be operated during application of random vibration so that the functional effects caused by these tests may be evaluated. The amplifier shall meet performance requirements, as specified, while the functional vibration levels are being applied and immediately preceding and following the application of the endurance levels.

4.4.4.2 Mounting techniques. The amplifier shall be mounted directly to the vibration exciter or transition table by means of a MT-6330/ARC mount with F-1556/ARC Filter and RT-1250A/ARC Receiver-Transmitter in the normal system mounting configuration. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the amplifier. Additional vibration sensors shall be located in or on the amplifier to determine resonant frequencies and amplification factors. Locations to be selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal. The amplifier shall be mounted on the MT-6330/ARC mount with other WRA's or equivalent weights.

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4.4.4.3 Performance of test. The individual amplifier shall be subjected to broadband random vibration excitation with the power spectral density envelope shown on Figure 3. The amplifier shall be attached to the vibration exciter in accordance with 4.4.4.2. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item. Two test levels are required, a functional level and an endurance level. For each axis, one half of the functional test shall be conducted first, then the endurance test, followed by the second half of the functional test. The amplifier shall perform according to the operating requirements as specified in 4.4.4.1. The acceleration power spectral density (g^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be as specified in 4.4.4.4. Test times shall, for each axis, be one hour each for functional and endurance levels. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100, -30 percent (+3, -1.5 dB) below 500 Hz and +100, -50 percent (+3 dB) between 500 Hz and 2,000 Hz except that deviations as large as +300, -75 percent (+6 dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum, between 500 and 2,000 Hz. Tolerance levels in terms of dB are specified as:

$$dB = 10 \text{ Log}_{10} \frac{W1}{W0}$$

Where W1 = measured acceleration power spectral density in g^2/Hz units, the term W0 defines the specified level in g^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT=50 minimum. Specific analyzer characteristics shall be as specified below or equivalent, subject to the BT=50, minimum limitation.

a. On-line, contiguous filter, equalization/analysis system having a bandwidth as follows:

B=25 Hz, maximum between 20 and 200 Hz
B=50 Hz, maximum between 200 and 1,000 Hz
B=100 Hz, maximum between 1,000 and 2,000 Hz

b. Swept frequency analysis systems characterized as follows:

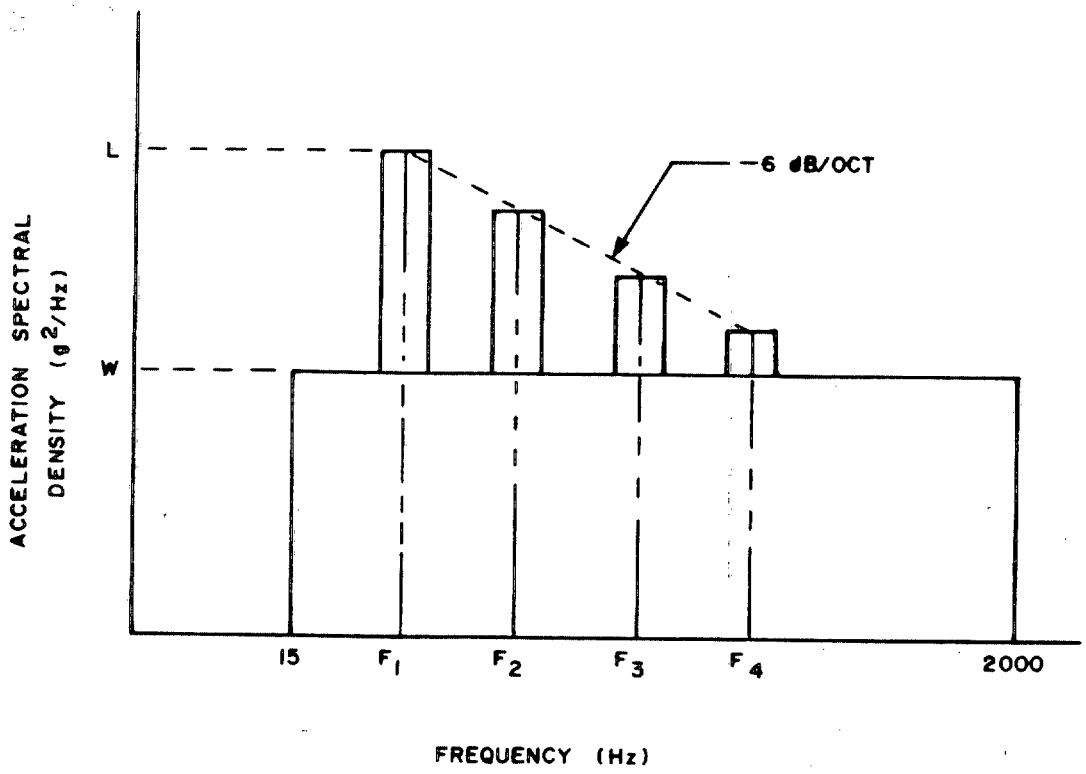
(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B=25 Hz, maximum between 20 to 200 Hz
B=50 Hz, maximum between 200 to 1,000 Hz
B=100 Hz, maximum between 1,000 to 2,000 Hz

(b) Analyzer averaging time =T= 2 RC = 1 second, where T= true averaging time constant

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L SEE 4.4.4.4

W SEE 4.4.4.4

F_1 = 68 Hz, $F_2 = 2F_1$, $F_3 = 3F_1$, $F_4 = 4F_1$

BANDWIDTH AT EACH Fi= ± 5% Fi

OVERALL LEVEL = 2.45 grms

FIGURE 3. Random vibration envelope (see 4.4.4.3)

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c) Analysis sweep rate (linear) =

$$R = \frac{B}{4RC} \text{ or } \frac{B^2}{8} \quad \text{(Hz/second)}$$

(2) Constant percentage bandwidth analyzer

(a) Filter bandwidth - pfc=one-tenth of center frequency maximum (0.1 fc) where p=percentage and fc=analyzer center frequency

(b) Analyzer average time = T = $\frac{50}{pfc}$, minimum

(c) Analysis sweep rate (logarithmic) =

$$T = \frac{pfc}{4RC} \text{ or } \frac{(pfc)^2}{8}$$

(Hz/second), maximum, whichever is smaller

Digital power spectral density analysis employing quantization techniques providing accuracies corresponding to the above approach.

4.4.4.3.1 Worst case axis. During the vibration testing, data shall be obtained to identify the worst case axis for purpose of vibration screening tests on production equipments.

4.4.4.4 Test levels. For the purpose of the random vibration test, the following test levels shall be used:

a. Functional Level-W = $0.01g^2$/Hz, L = $0.3g^2$/Hz (4.9g rms minimum)

b. Endurance Level-W = $0.2g^2$/Hz, L = $0.6g^2$/Hz (6.9g rms minimum)

4.4.5 Reliability qualification test. A reliability qualification test may be included in the first article tests. A maximum of six amplifiers shall be tested in accordance with MIL-STD-781 using the reliability qualification test - mission profile in this specification. (See Figures 4 and 5.) Each amplifiers shall have successfully completed the normal burn-in and all equipment test (AET) prior to the start of RQT. The mission profile test cycle should begin at the "12 hour" point which is a cold day ground stabilization (-54°C) prior to ground operation and flight in a cold day. Random vibration should be applied with the units mounted on the MT-6330/ARC mount with other WRA's or equivalent weights. The accept-reject criteria imposed shall be those of MIL-STD-781, Test Plan IIIC, with the upper test MTBF specified in 3.4.1.3.

4.4.6 Maintainability demonstration. A maintainability demonstration for intermediate level requirements shall be conducted in accordance with MIL-STD-2084 and MIL-STD-471. Test Method 1, Test Plan B of MIL-STD-471, shall be used.

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4.4.7 Production amplifiers. Amplifiers supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be capable of successfully passing the same tests as specified herein.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing may be under the supervision of the procuring activity representative. The contractor shall retain test data showing quantitative results for all quality conformance tests. Such tests shall be signed/stamped by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual tests (see 4.5.1)

b. Manufacturing screening (see 4.5.2)

c. Sampling tests (see 4.5.3)

d. Reliability assurance tests (see 4.5.4)

e. Special tests (see 4.5.5)

4.5.1 Individual tests. Each amplifier submitted for acceptance shall be subjected to the individual tests. The tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each amplifier accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

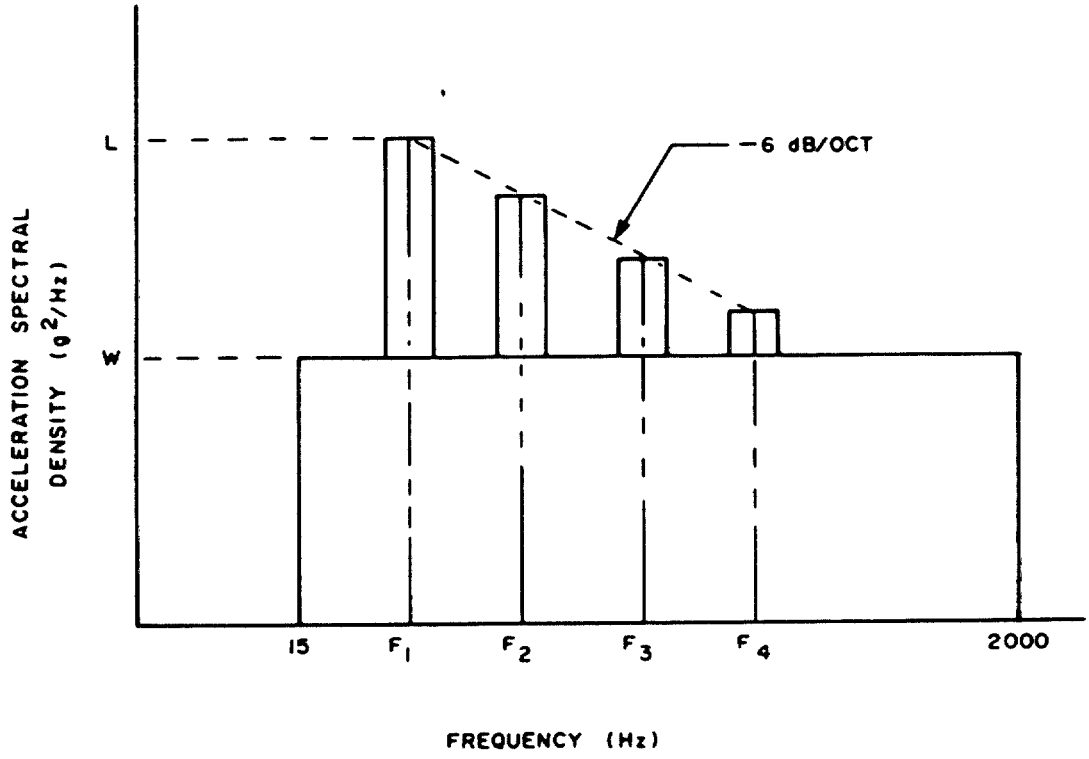
b. Operational test (see 4.5.1.2)

4.5.1.1 Examination of product. Each amplifier shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each amplifier shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure specified equipment operation.

4.5.2 Manufacturing screening. The contractor shall subject each amplifier to manufacturing screening prior to undergoing sample tests or reliability assurance tests. Manufacturing screening shall consist of random vibration and temperature cycling. An additional screening period may be used at the option of the contractor. If an additional screening period is to be used, the details thereof shall be included in the approved test procedures and shall be the same for all equipment, including spare units.

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L = .075 g^2/Hz

W = .0025 g^2/Hz

F_1 = 68 Hz, $F_2 = 2F_1$, $F_3 = 3F_1$, $F_4 = 4F_1$

BANDWIDTH AT EACH Fi= ± 5% Fi

OVERALL LEVEL = 2.45 grms

FIGURE 4. RQT vibration profile (see 4.4.5)

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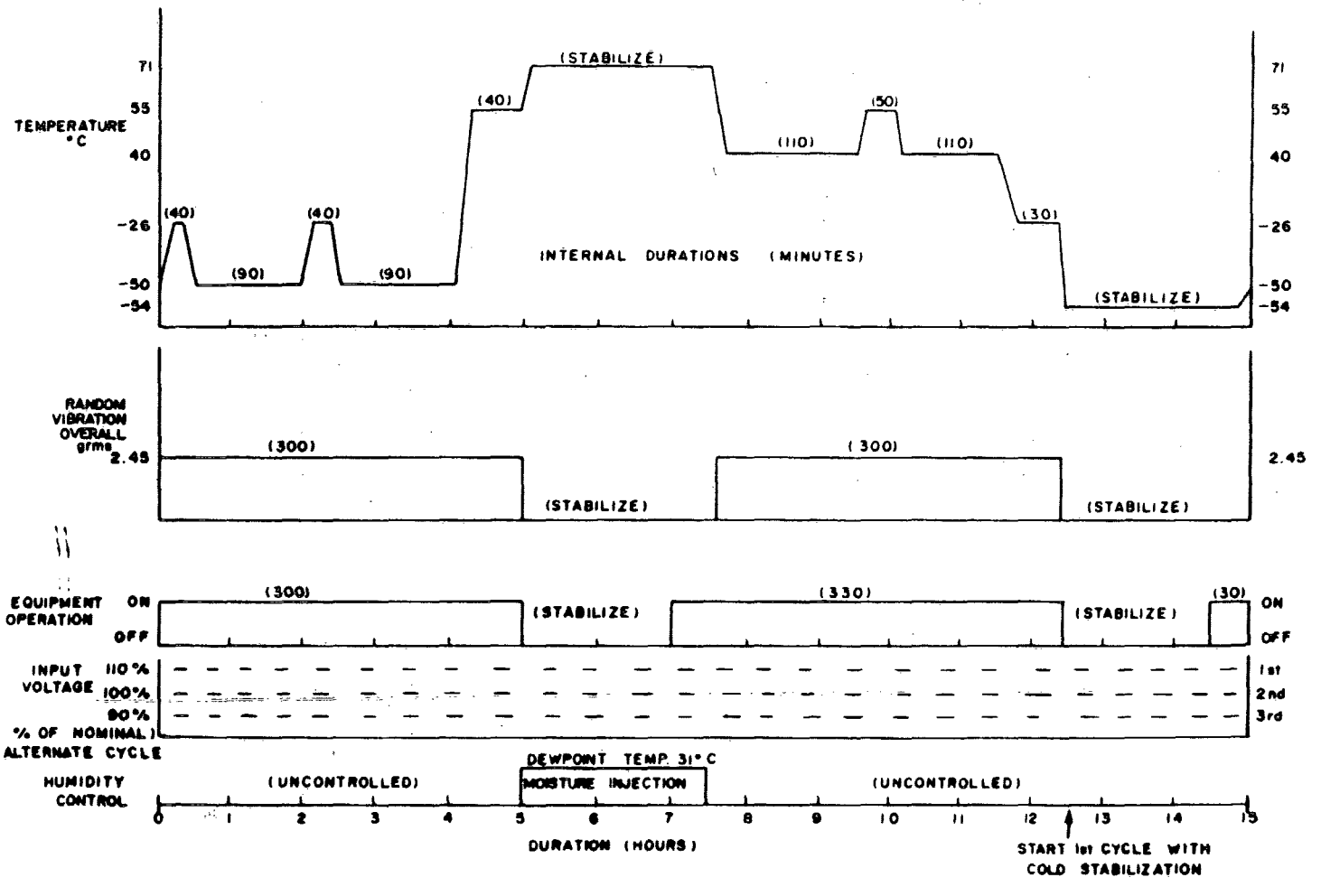


FIGURE 5. Environmental cycle (ref. 4.4.5)

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4.5.2.1 Random vibration. Prior to conducting temperature cycling, each amplifier shall have completed a 10-minute failure-free vibration screen at 3.3g rms. Power spectral density shall cover 20-2000 Hertz. Pseudo-random vibration methods are acceptable. The amplifier shall be tested as part of a system while mounted on an MT-6330/ARC, or equivalent, which shall be hard mounted to the vibration source without vibration isolators. The vibration envelope is shown on Figure 6.

4.5.2.2 Temperature cycling. Each amplifier shall undergo 25 temperature cycles between -54°C and 55°C. The rate of temperature change between extremes shall be at least 5°C per minute. When the amplifier temperature has stabilized within 5°C of either temperature extreme, the amplifier shall be soaked. Hot soak at +55°C shall be two hours; cold soak duration at -54°C shall be one hour. At the conclusion of each soak, the next temperature ramp shall begin. The amplifier shall be energized throughout temperature cycling except during the cooldown period from +55°C to -54°C and during the 60 minute soak at -54°C. In addition, the amplifier shall be turned off for one minute at points 30, 60, and 90 minutes into each hot soak period. Amplifier performance shall be periodically checked during each cycle and repairs shall be made to correct failures. The last five consecutive temperature cycles shall be failure free. One temperature cycle is defined as: operation at +55 C for the 120 minute hot soak (including three minutes "off"), non-operational cooldown to -54°C, 60 minute non-operational cold soak at -54°C, operational temperature increase to +55°C.

4.5.3 Sampling tests. Amplifiers shall be subjected to sampling tests. Amplifiers selected for sampling tests shall first have passed the individual tests. Amplifiers will be selected for sampling tests by the government inspector in accordance with Table VII.

TABLE VII. Sampling test sample selection

Quantity of equipments offered for acceptance	Quantity to be selected for sampling test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when reliability assurance tests are conducted.

4.5.3.1 Scope of sampling tests. As a minimum, each amplifier selected for sampling tests shall be subjected to the following tests:

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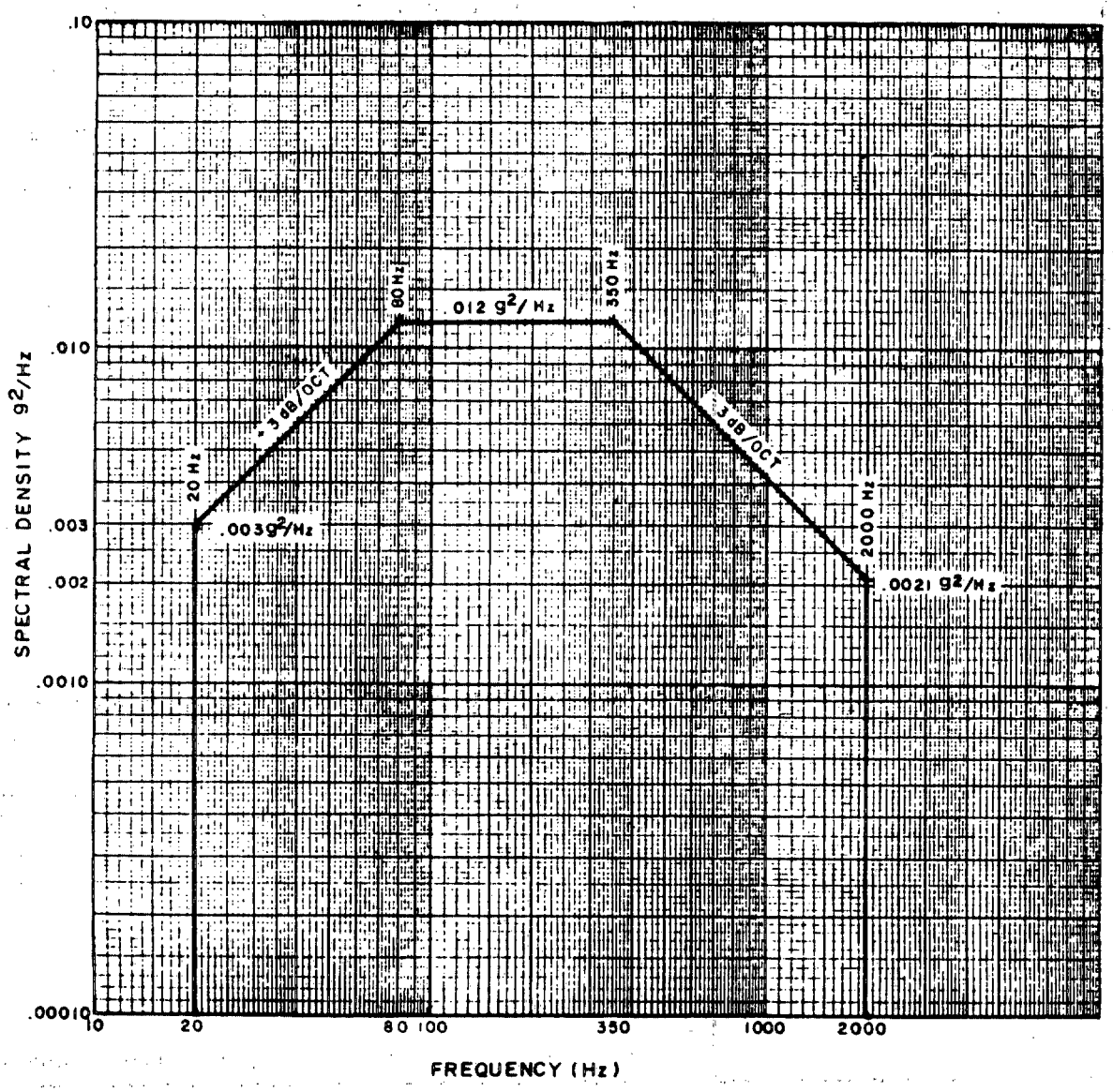


FIGURE 6. Manufacturing screening random vibration envelope (see 4.5.2.1)

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a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each amplifier tested and should be based on results of the first article, individual and special tests.

c. During the reliability sample test specified in 4.5.4.2, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurring in future equipment taken.

4.5.4 Reliability assurance tests. A reliability assurance test shall be conducted using MIL-STD-781. Classification of failure shall be in accordance with 4.5.7.

4.5.4.1 All equipment test. Each amplifier shall be tested as specified in MIL-STD-781 and modified herein. Environmental cycling shall be as shown on Figure 5. No vibration is required. Each amplifier shall have successfully completed manufacturing screening as specified in 4.5.2 prior to the start of the test. Each equipment (WRA) shall be tested for 125 equipment operating hours with the last two cycles failure free. For determining MTBF compliance each amplifier shall be tested separately as opposed to testing as an equipment system. The test MTBF shall be as listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating time and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan XVIIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan XVIIIC curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of amplifier at any time that a reject situation exists pending a review of the contractors efforts to improve the equipment, the equipment parts, or the equipment workmanship so that the entire compilation will show other than a reject decision. The amplifier shall be mounted on MT-6330/ARC Mount with other WRA's or equivalent weights.

4.5.4.2 Reliability sample test. When specified in the contract (see 6.2.1) a monthly sample test of amplifiers shall replace the reliability test of 4.5.4.1. Units shall be tested as specified in MIL-STD-781 as modified herein. Temperature cycling and power on/off cycling are required and shall be as shown on Figure 5. Each sample amplifier shall be tested for 256 hours with the last two cycles failure free. For determining MTBF compliance each designated equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The MTBF for each designated equipment shall be those listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan IIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. The sample size shall be based on the production quantity of that equipment for the month as specified in Table VIII.

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TABLE VIII. Reliability test sample size

Production Quantity	Sample Size
2 - 5	1
6 - 25	2
26 - 50	3
51 - 100	4

4.5.4.3 Test details. Test details such as the performance criteria to be measured, special failure criteria, and preventive maintenance to be allowed during the test shall be part of the test procedures to be prepared by the contractor prior to the beginning of the reliability assurance tests.

4.5.5 Special tests. When specified in the contract (see 6.2.1), special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the amplifier and to assure quality control. The amplifier selected for special tests may be selected from amplifiers previously subjected to the sampling or reliability tests.

4.5.5.1 Special test schedule. Selection of amplifiers for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.5.2 Scope of special tests. Special tests shall consist of such tests as authorized by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available to the procuring activity for acceptance prior to conducting the tests.

4.5.6 Equipment failure. Should a failure occur during either the sampling, reliability qualification, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure and the equipment BIT performance in detecting and isolating the failure.

b. Determine if the failure is an isolated case or design defect or pattern failure.

c. Prepare proposed corrective action intended to reduce the possibility of the same component or BIT failure(s) occurring in future tests or documented justification where no corrective action is proposed.

d. Where practical, include a test in the individual test to check all amplifiers for this requirement until assurance is obtained that the defect has been corrected.

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4.5.7 Failure classification. All incidents occurring during reliability assurance tests shall be classified and reported as either relevant or non-relevant in accordance with MIL-STD-2074 and MIL-STD-781. Only those incidents classified as relevant failures shall be accountable for making an accept/reject decision.

4.5.7.1 Relevant failures. All test incidents shall be considered as relevant failures unless determined to be non-relevant in accordance with the criteria of MIL-STD-2074.

4.5.7.2 Non-relevant failures. Although non-relevant failures are not used for reliability assurance test accept/reject decisions, all test article incidents and equipment failures shall be recorded and reported.

4.6 Test procedures. The procedures used for conducting all tests required by Section 4 shall be prepared by the contractor. The right is reserved by the procuring activity or the government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of the specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures.

4.7 Testing of spares and repair parts. Equipment delivered as spare or repair parts shall undergo acceptance tests to ensure that the items meet performance and reliability requirements not less than those imposed in production.

4.7.1 Weapons replaceable assemblies. All items to be delivered as spares at the weapons replaceable assembly (WRA) or higher level shall receive performance, manufacturing screening and reliability assurance tests commensurate with that afforded the original production equipment.

4.7.2 Shop replaceable assemblies. All items to be delivered as spares at the shop replaceable assembly (SRA) or sub-SRA level shall receive functional checks and manufacturing screening under conditions and for a duration commensurate with that imposed on that assembly in the original production.

4.7.3 Repair parts. All electronic component parts to be delivered as repair parts shall receive functional and screening tests equivalent to those imposed for the nearest military standard part.

4.7.4 Test details. The details of acceptance and screening tests to be performed on SRA's and WRA's supplied as spare or repair parts shall be incorporated into item specifications, drawings and test procedure.

4.8 Presubmission testing. No item, part, or complete equipment shall be submitted until it has been previously tested by the manufacturer and found to comply, with all applicable requirements of this specification and the contract.

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5. PACKAGING

5.1 General. All major units and parts of the amplifier shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555 for the level of shipment specified in the contract or purchase order (see 6.2.1).

5.2 Marking. Shipments shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The AN/ARC-182(V) System is a variable combination of units which includes the AM-7177/ARC Radio Frequency Amplifier that provides UHF/VHF AM and FM voice and data communication in tactical aircraft. The AM-7177/ARC is a UHF amplifier intended for use to increase the system UHF output level from 10 to 30 watts. The AM-7177/ARC Amplifier is designed for attaching to a MT-6330/ARC Mounting and operating with the RT-1250A/ARC Receiver-Transmitter, or equivalent and the F-1556/ARC UHF Filter.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. If a first article inspection is required (see 3.2.)

c. If samples of non-standard parts and materials are not required (see 3.3.5)

d. If reliability program is not to be established by contractor (see 3.4.1)

e. If a monthly reliability sample test is to be conducted (see 4.5.4.2)

f. Test procedure modifications or additional tests, if required (see 4.5.5, 4.6)

g. Selection of applicable levels of packaging and packing (see 5.1)

h. Whether All Equipment Test or Reliability Sample Test is required (see 4.5.4.1, 4.5.4.2)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD Far Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked, and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

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Paragraph	Data requirement title	Applicable DID no.
3.3.2	Engineering Change Proposals (ECPs) and Requests for Deviations and Waivers (Long Form)	DI-E-2037
3.3.3	Program Parts Selection List (PPSL)	DI-MISC-80072
3.4.1.2	Part Approval Request	DI-MISC-80071
3.4.8	Request for Confirmation of Nomenclature	DI-CMAN-80194
	Request for Assignment of Serial Number and Serial Number Prefix Letters	DI-CMAN-80195
	Request for Assignment of Nomenclature	UDI-E-21582
	Request for Approval of Identification Plate Drawings	DI-CMAN-80196
4.4, 4.6	First Article Inspection Procedure	DI-T-4901
4.4.2	First Article Inspection Report	DI-T-4902
4.4.3.1, 4.4.3.2, 4.4.3.5	Electromagnetic Interference Test Plan	DI-EMCS-80201
4.4.3.3	Electromagnetic Interference Test Report	DI-EMCS-80200
4.5.4.1, 4.5.4.2	Reliability Test Report	DI-RELI-80252
4.5.4.3	Reliability Test Procedures	DI-RELI-80251
4.5.5.2	Production/Acceptance Inspection Procedures	DI-T-4903
4.5.6c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719
4.5.7.2	Failed Item Analysis Report	DI-RELI-80253
4.7.4	Drawings, Engineering and Associated Lists	DI-E-7031

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed a such in DOD 5010.12L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 Options. The following options are available:

a. An additional manufacturing screening period is allowable at the option of the contractor (see 4.5.2).

b. Acceptance of equipment may be stopped anytime a reject situation exists in the All Equipment Test. This is at the option of the procuring activity (see 4.5.4.2).

6.4 First article. When a first article inspection is required the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Associated equipment. The amplifier shall operate with the associated equipment listed in Table IX which shall not be supplied as part of the amplifier:

TABLE IX. Associated equipment

Item Designation	Equipment	Military Specification/ Naval Air Systems Command Drawings
Receiver-Transmitter	RT-1250A/ARC	MIL-R-85664
Control, Radio Set	C-10319A/ARC	MIL-C-85666
	C-10776A/ARC	MIL-C-85666
	C-11131/ARC	MIL-C-85666
Mounting, Radio Set	MT-6330/ARC	MIL-M-85677
Filter, Radio Frequency	F-1556/ARC	MIL-E-85675
Control, Radio Set	C-11128/ARC	MIL-C-85806
	C-11129/ARC	MIL-C-85806
	C-11130/ARC	MIL-C-85806

6.6 Subject term (key word) listing.
Amplifier
AN/ARC-182(V) Radio Set
Built-In-Test (BIT)
Linear Amplification
Radio Frequency (RF)
Remote Mount
Thirty (30) Watts
Ultra High Frequency (UHF)

Preparing Activity:
Navy AS
(Project 5821-N192)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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